Quiz 01 Weeks 01 and 02

Teaching Week 03 | Sample questions with the answers marked, for study.

Name _____

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Circle one letter for each question. Each question has exactly one correct answer. Course notes and your own notes are permitted. Generative AI services and communication with other people are not permitted.

1. In CRISP-DM, which stage cleans and transforms the data so that it is ready for modelling?

A. Business understanding
B. Data understanding
C. Data preparation
D. Evaluation
E. Deployment

2. In an $n \times p$ data matrix, what does each of the n rows represent?

A. An observation, also called a case or record
B. A variable measured on every case
C. A summary statistic of one column
D. A category of a nominal variable
E. A missing-value code

3. For $x = 2, 4, 4, 5, 7, 9, 12, 41$, what is the median?

A. 5 **B. 6** C. 7 D. 10.5 E. 12

4. The boxplot rule flags an observation as a possible outlier when it lies outside which interval?

A. $[\,\bar{x} - s,\ \bar{x} + s\,]$
B. $[\,\min,\ \max\,]$
C. $[\,Q_1,\ Q_3\,]$
D. $[\,Q_1 - 1.5\,IQR,\ Q_3 + 1.5\,IQR\,]$
E. $[\,\text{median} - IQR,\ \text{median} + IQR\,]$

5. In the right-skewed examples from class, how are the mean, median, and mode ordered?

A. mean < median < mode
B. median < mode < mean
C. mode < mean < median
D. All three are equal
E. mode < median < mean

6. Which set of rules describes a tidy table?

A. One variable per column, one observation per row, one value per cell
B. Observations stored in the column names
C. One column for each distinct value
D. Every incomplete row removed
E. Rows and columns sorted alphabetically

7. High earners leave a survey's income question blank more often, and no recorded variable explains the pattern. Which mechanism is this?

A. MCAR, missing completely at random
B. MAR, missing at random
C. Complete-case deletion
D. MNAR, missing not at random
E. Available-case analysis

8. Two missing entries are filled with the observed mean. What happens to the mean and the standard deviation (SD) of the filled column?

A. The mean rises; the SD is unchanged
B. The mean and the SD are both unchanged
C. The mean falls; the SD rises
D. The missing values are recovered exactly
E. The mean is unchanged; the SD decreases

9. A variable is standardized as $z = (x - \bar{x})/s$, using its observed mean and sample standard deviation. The result has

A. its original units and its original spread
B. every value between 0 and 1
C. mean 0 and standard deviation 1, with no units
D. its outliers removed
E. a symmetric distribution

10. How does a stratified sample differ from a simple random sample?

A. It keeps every record in the table
B. It groups the records, then samples within each group
C. It repairs a biased source dataset
D. It takes the first rows of the table
E. It makes R and Python samples identical

SAMPLE QUESTIONS